

Augmentation and leakage

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training a small image model

Without opening the old lesson, explain one key idea from Training a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Keep the equality boundary for Augmentation and leakage. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Augmentation and leakage. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Design a grouped split independently

Six fictional source groups J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to training, M to validation and N, O to final test. Keep all four records per training group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Check the label after a transformation

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

My reasoning and evidence

6. W2 Keep the equality boundary

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

My reasoning and evidence

7. W3 Audit related records

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original. Identify the overlap and explain what must be rerun if this validation set already selected a model.

My reasoning and evidence

8. W4 Design a grouped split independently

Six fictional source groups J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to training, M to validation and N, O to final test. Keep all four records per training group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My reasoning and evidence

9. W5 Match a training budget

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches. Give both update counts and one way to match budgets without using final outcomes.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled augmentation and leakage evidence sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Augmentation and leakage

Keep every augmented image with its original source group.

Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Augmentation and leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Track provenance

Each flipped copy retains its source ID and parent image reference.



2. Assign by source

A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.



3. Check label meaning

Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

CHANGE ONE THING

Move D's flipped copy into training.

Explain the changed result using the matching lesson.

How many independent source groups are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep every augmented image with its original source group.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

1. Assign by source
2. Track provenance
3. Check label meaning

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check label meaning Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign source groups to training, validation and final evaluation before producing copies. Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Practice 2: explain the deciding evidence

Changed-case practice: Move D's flipped copy into training. Which conclusion is supported by the supplied facts?

1. Final-test group separation fails.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Repair a missing building block

Review Training a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower training loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
